



## Unit-1 (BME)

**Introduction to Mechanical Engineering:** Role of Mechanical Engineering in Industries and Society – Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors.

**Engineering Materials-** Metals- Ferrous and Non-ferrous, Ceramics, Composites, Smart Materials.

### **What is Mechanical Engineering?**

Mechanical Engineering is defined as the branch of engineering that deals with the design, development, construction, and operation of mechanical systems and tools. It includes machines, tools, and equipment used in various industries, such as transportation, manufacturing, power generation, and medical devices etc.

### **Role of Mechanical Engineering:**

Mechanical engineers improve the global supply chain and scale up therapeutics in addition to helping with production. The digital world of the pandemic integrated with the wave of new technologies from the fourth industrial revolution as engineers remotely conceived designs.

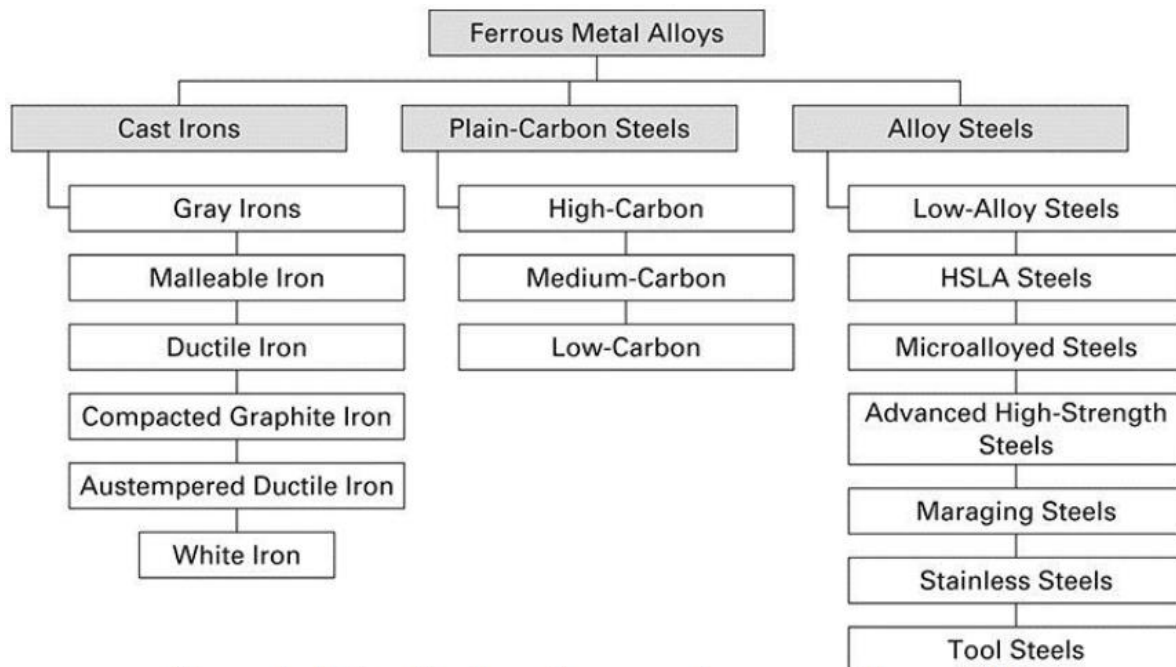
- Mechanical engineers are increasingly involved in the design, development, manufacturing, and testing of hybrid electric vehicles, battery management systems, and electrical safety as electric vehicle technology advances in the automotive industry.
- In recent years, they have made significant contributions to the development of renewable energy infrastructure around the world, ranging from large solar projects to using wind, ocean, or tidal energy to generate power.
- Power Generation: Mechanical engineers design and develop power-generating machines such as internal combustion engines, gas turbines, and steam and wind turbines etc.
- Heating and Cooling Systems: They design and develop heating, ventilation, refrigeration and air conditioning systems for buildings and other structures.
- Transportation: Mechanical engineers are involved in designing and developing transportation systems, including cars, trains, airplanes, steamers and boats.
- Industrial Equipment: They design, develop and maintain industrial equipment such as machine tools, robots, and conveyor systems & belts.
- Infrastructure: Mechanical engineers play a key role in the design and maintenance of infrastructure, including buildings, bridges, roads, and transportation systems.
- Mechanical Engineers are involved in designing, building, and maintaining the engines, machines, and structures that make modern life possible and comfortable.

## Ferrous materials:

Ferrous means iron. Iron is the name given to pure ferrite Fe, as well as to fused mixtures of this ferrite with large amount of carbon (may be 1.8%), these mixtures are known as pig iron and cast iron. Primarily pig iron is produced from the iron ore in the blast furnace from which cast iron, wrought iron and steel can be produced.

Ferrous materials/metals may be defined as those metals whose main constituent is iron such as pig iron, wrought iron, cast iron, steel and their alloys. The principal raw materials for ferrous metals are pig iron. Ferrous materials are usually stronger and harder and are used in daily life products.

*Classification of general ferrous metals and alloys:*



## Properties:

- Ferrous metals are good conductors of heat and electricity, making them valuable in electrical appliances and for carrying an electric current over a distance with little energy lost.
- Metal alloys have high resistance to shear, torque and deformation.
- The thermal conductivity of metal is useful for containers to heat materials over a flame.

## Applications:

- ❖ General engineering
  - Aircraft construction
  - Electric conductors
  - Building construction



- ❖ Railroad tracks and signs
- ❖ Automobile industry
- ❖ Use in high rise building and bridge construction

### **Non-ferrous materials:**

Non-ferrous metals or alloys are defined as materials that are not iron-based like their ferrous counterparts. Common non-ferrous metals include copper, aluminium, lead, tin, zinc, gold, silver etc. and their alloys.

The simplest way to see if a metal is non-ferrous is to use a magnet. If the magnet does not stick, the metal is non-ferrous – meaning, without iron.

### **Properties:**

- High corrosion resistance.
- Good thermal and electrical conductivity.
- Low density.
- Non-magnetic.
- Colourful.
- Easy to fabricate (including machinability, casting, and welding)

### **Applications:**

| <b>Material/alloy</b>    | <b>Properties</b>                                                                                                                                                   | <b>Applications</b>                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Al and its alloys        | <ul style="list-style-type: none"><li>• Nonmagnetic</li><li>• Light in weight</li><li>• Have good formability</li><li>• Excellent weight-strength ratio</li></ul>   | <ul style="list-style-type: none"><li>• Cryogenic vessels</li><li>• Chemical vessels</li><li>• Aircraft and aerospace industry</li><li>• Ships and vehicles</li><li>• Power plants etc.</li></ul> |
| Copper and copper alloys | <ul style="list-style-type: none"><li>• Good corrosion resistance</li><li>• Machinability</li><li>• Good for cold working</li></ul>                                 | <ul style="list-style-type: none"><li>✓ Fasteners</li><li>✓ Hydraulic tubing</li><li>✓ Radiators</li><li>✓ Bearings and valve guides</li></ul>                                                    |
| Nickel and nickel alloys | <ul style="list-style-type: none"><li>• Good corrosion and oxidation resistance</li><li>• used in high-temperature applications in corrosive environments</li></ul> | <ul style="list-style-type: none"><li>❖ Air craft gas turbine</li><li>❖ Medical instruments</li><li>❖ Automotive spark plugs</li><li>❖ Nuclear power plants</li></ul>                             |



|                              |                                                                                                                                                                                  |                                                                                                                                                      |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Titanium and titanium alloys | <ul style="list-style-type: none"><li>• high melting point</li><li>• good corrosion resistance</li><li>• high strength-to-weight ratios</li><li>• good for hot working</li></ul> | <ul style="list-style-type: none"><li>• aero planes</li><li>• missiles</li><li>• rockets</li><li>• medical implants</li><li>• power plants</li></ul> |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

**Difference between ferrous and non-ferrous materials:**

| <b>Ferrous materials</b>                                                       | <b>Non-ferrous materials</b>                                                         |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Ferrous metals mostly contain iron as a main constituent.                      | Non-ferrous metals do not contain iron.                                              |
| Ferrous metals are magnetic.                                                   | Non-ferrous metals are non-magnetic                                                  |
| They give little resistance to corrosion                                       | They are usually more resistant to corrosion than ferrous metals                     |
| Ferrous metals include mild steel, carbon steels, cast iron, wrought iron etc. | Non-ferrous metals include copper, aluminium, nickel, tin, lead, zinc etc.           |
| Ferrous metals are heavier, have high strength and durability.                 | Non-ferrous metals are much higher and malleable than ferrous metals                 |
| They are suitable for building construction, rail-road, bridges, piping etc.   | They are suitable for electrical wiring, electronic components, heat exchangers etc. |

**Ceramic materials:**

Ceramics are compounds between metallic and non-metallic elements for which the inter-atomic bonds are either ionic or predominantly ionic.

Ceramics are defined as products made from inorganic materials having non-metallic properties, usually processed at a high temperature at some time during their manufacture.

**Properties:**

Ceramic products have a number of outstanding properties which determine their usefulness. One of the most unusual of these is their great durability. This durability can be divided into three types: **chemical, mechanical and thermal**.

- High strength and hardness
- High elastic module, High brittleness
- High dimensional stability.
- Good wear resistance.
- High resistance to corrosion and chemical attack, oxidation resistant.
- High weather resistance.
- High melting point, refractory and thermal insulators,



## Types of ceramics:

**Glasses:** these are non-crystalline silicates containing other oxides, usually  $\text{CaO}$ ,  $\text{Na}_2\text{O}$ ,  $\text{K}_2\text{O}$  and  $\text{Al}_2\text{O}_3$  which influence the glass properties and its colour. Containers, windows, mirrors, lenses, etc.

**Clay products:** clay is the one of most widely used ceramic raw material. It is found in great abundance and popular because of ease with which products are made. structural products: bricks, tiles, sewer pipes and white-wares: porcelain, chinaware, pottery, etc.

**Refractories:** these are described by their capacity to withstand high temperatures without melting or decomposing; and their inertness in severe environments. Thermal insulation is also an important functionality of refractories.

**Abrasive ceramics:** these are used to grind, wear, or cut away other material. Thus, the prime requisite for this group of materials is hardness or wear resistance in addition to high toughness. Diamond, silicon carbide, tungsten carbide, silica sand, aluminium oxide / corundum are some typical examples of abrasive ceramic materials.

**Cements:** cement, plaster of paris and lime come under this group of ceramics. The characteristic property of these materials is that when they are mixed with water, they form slurry which sets subsequently and hardens finally. Thus, it is possible to form virtually any shape. They are also used as bonding phase, for example between construction bricks.

**Advanced ceramics:** these are newly developed and manufactured in limited range for specific applications. Usually their electrical, magnetic and optical properties and combination of properties are exploited. Typical applications: heat engines, ceramic armours, electronic packaging, etc.

| Ceramic                                               | Application                                                                                                                                                    |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminium oxide / Alumina ( $\text{Al}_2\text{O}_3$ ) | <ul style="list-style-type: none"><li>insulators in spark plugs</li><li>dental and medical use</li><li>making lasers.</li></ul>                                |
| Aluminium nitride ( $\text{AlN}$ )                    | <ul style="list-style-type: none"><li>electrical circuits operating at a high frequency</li></ul>                                                              |
| Diamond (C)                                           | <ul style="list-style-type: none"><li>industrial abrasives</li><li>abrasion resistant coatings</li><li>cutting tools</li><li>jewelry</li></ul>                 |
| Silica ( $\text{SiO}_2$ )                             | <ul style="list-style-type: none"><li>laboratory glassware</li><li>communications media (optical fiber)</li><li>thermal insulation</li><li>abrasives</li></ul> |
| Silicon carbide ( $\text{SiC}$ )                      | <ul style="list-style-type: none"><li>used as semiconductor</li><li>automotive and gas turbines</li></ul>                                                      |



## Composite materials:

A composite material may be defined as an artificially prepared or natural multiphase material that exhibits a significant property of the both the constituent material such as high strength, stiffness and high coefficient of thermal expansion in which the chemically dissimilar phases are separated by distinct interface.

### Properties:

- high specific strength and stiffness
- excellent resistance to fatigue and creep
- good corrosion and wear resistance
- low coefficient of thermal expansion
- low density

### Constituents of Composites

Two essential constituents of composites are

I. *Matrix phase*: It is the continuous body constituent (Dispersion phase) which encloses the composite and gives its bulk form. It may be polymer, metal or ceramic material.

II. *Dispersed phase*: It is the Structural constituent (Dispersed phase) which determines internal structure of the composite and gives its bulk form. It may be Fiber, Particulate, Flakes or Whiskers.

### Types of composite materials:

Based on the dispersed phase in the given matrix of composite they are classified as

I. Fiber reinforced Composite II. Particulate Composite III. Structural Composite

#### *I. Fiber reinforced Composite*

- It is Consist of dispersed phase fiber and a continuous or dispersion phase polymer or metal or metal alloy with a bonding agent. The fiber can be employed in the form of continuous length, staples or whiskers.

- Such composites possess high specific strength, specific modulus, stiffness, corrosion resistance and lowers density

- The mechanical characteristics of FRC depend on the following

1. Properties of fiber
2. Interfacial bond between fiber and matrix
3. Fiber length like longer gives continuous, shorter length gives discontinuous or random.

Reinforcement efficiency of continuous is higher than short fibers.





4. Fiber orientation and concentration i.e. if it is orderly orientation and continuous it is highly anisotropic or discontinuous or random orientation

*limitations:* High cost

*Applications:* Structural components of air craft like wings and bodies, sport equipment, fishing rods etc.

3). Alumina oxide/ carbon fiber reinforced metal composites:

Fibers of alumina or carbon dispersed in metal or metal alloy matrix which possess improved specific strength, stiffness, wear resistance, creep resistance and resistance to thermal distortion etc.,. Ex: i. Fiber  $\text{Al}_2\text{O}_3$ /carbon in a matrix metal alloy find applications in the preparation of components of automobile engines.

ii. Fiber  $\text{Al}_2\text{O}_3$  / $\text{W}_2\text{O}_3$  in a matrix of Ni or Co based alloy find applications in the preparation of components of turbine engines.

### *II. Particulate Composite*

The solid particulates of metal oxides or carbides of varying size and form dispersed in metal, metal alloy, ceramic or polymer liquid matrix.

*Particle reinforced composites are further classified into the following two types*

i) Large -particulate composites ii) Dispersion strengthened composites

### *III. Structural composites*

Structural composites are prepared by Compressing the stacking of layers of fiber reinforce composites

These are of two types 1. Laminated composites 2. Sandwich composites.

Example: Plywood, Copper bottom steel articles

### **Applications:**

- ✓ automobiles industries: Automobile parts like components of engine, spray nozzle, mud guards, tires etc.
- ✓ Aeronautical applications: structural components like wings, body & stabilizer and fuel of aircraft, rocket army missiles in military etc.
- ✓ Marine applications: shaft, hulls, spars and other part of ships
- ✓ Safety equipment like helmets
- ✓ Sport equipment like tennis rockets, golf sticks, other safety equipment
- ✓ Communication Industry like preparation of antennae and electronic circuit boards



## **Smart Materials:**

Smart materials, also called intelligent or responsive materials, are designed materials that have one or more properties that can be significantly changed in a controlled fashion by external stimuli, such as stress, moisture, electric or magnetic fields, light, temperature, pH, or chemical compounds.

Smart Materials are categorized on the basis of their properties such as Active and Passive. Active Smart Materials possess the capability of modifying their geometric and material properties under the application of electric, thermal or magnetic fields there by acquiring an inherent capacity to transduce energy.

## **Properties:**

They have some amazing properties that distinguished them from other materials:

- Transiency: they can respond to various types of external stimulus.
- immediacy: the response time very less.
- self-actuation: this is the ability to change their appearance and shape.
- selectivity: the response is divided and expected.

## **Types of smart materials:**

- Optical fibers
- Piezo polymers
- Nano materials
- Shape memory alloy
- Self-healing materials
- Bi-component fiber

## **Applications:**

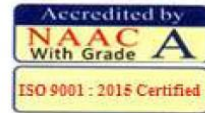
- ✓ Sensors and actuators
- ✓ (Medical implantation) Artificial muscles
- ✓ Nano engineering systems
- ✓ In drug delivery system
- ✓ As reversible bio catalyst
- ✓ In textile engineering
- ✓ As Glucose sensor
- ✓ In oil recovery





DVR & Dr. HS  
**MIC College of Technology**  
**AUTONOMOUS**

(Approved by AICTE & Permanently Affiliated to JNTUK, Kakinada)  
Kanchikacherla-521180, NTR Dist, A.P, India.  
Tel.No : 08678-273535/9491457799/7382616824  
Website : [www.mictech.edu.in](http://www.mictech.edu.in)



| Function                  | Definition                                                           | example                       | Application                                                         |
|---------------------------|----------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------|
| Actuators                 | Moves in response to an external stimuli                             | Shape-memory alloy            | Arterial stents                                                     |
| Colour changing           | Changes colour in response to an external stimuli                    | Photochromic materials        | Glasses that change colour when in direct sunlight                  |
| Self-healing              | A material that can repair itself without human influence            | Self-healing paint            | Kawasaki motorcycles repairing certain types of paintwork scratches |
| Temperature-responsive    | Materials that adapt to changes in temperature                       | Thermochromic thermometers    | Identifies whether a specific part of a machine is too hot to touch |
| Magnetorheological fluids | Changes viscosity with the application or removal of magnetic fields | (MR) Magnetorheological fluid | Reinforcing Kevlar armour                                           |